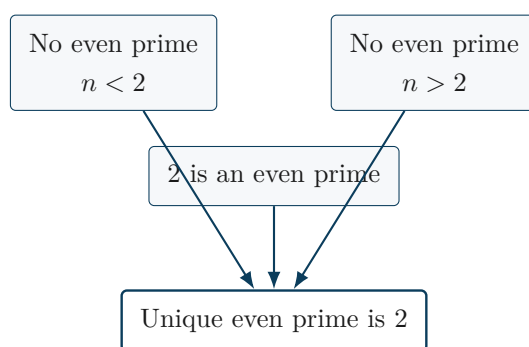

On the Uniqueness of the Even Prime

Five Elementary Proofs, a Two-Sided Strengthening,
and a Machine-Checked Formalization

A short note in elementary number theory
with a companion development in LEAN 4



Compiled from a LEAN 4 formalization (stdlib only, no Mathlib).
All theorems stated below have been checked by the LEAN kernel.

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Abstract

We give a careful elementary treatment of the classical fact that 2 *is the unique even prime number*. Five distinct proofs that there is no even prime greater than 2 are presented, together with the complementary observation that there is no even prime less than 2. From this expansive, two-sided picture we recover the grand uniqueness statement. We then discuss the specification gap inherent in interactive theorem proving—the distinction between verifying a formal derivation and validating that a formal statement matches an informal claim—and record how a companion LEAN 4 development closes as much of that gap as is possible inside the prover.

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1 Introduction

Among the first nontrivial observations one meets in elementary number theory is that evenness and primality interact in a very rigid way: once a natural number is divisible by 2 and strictly larger than 2, it cannot be prime. Consequently the only even prime is 2 itself.

The argument is short enough to fit in a sentence, yet rich enough to illustrate several standard proof patterns—direct reasoning from the definition, contraposition, contradiction, an appeal to the uniqueness of prime divisors, and a least-counterexample (well-ordering) argument. The same fact also admits a clean “two-sided” packaging: there are no even primes below 2 and none above 2, while 2 itself is an even prime. Trichotomy on $\mathbb{N}$ then yields uniqueness at a stroke.

The purpose of this note is twofold. First, we write the mathematics cleanly, with explicit definitions and several independent proofs. Second, we record how the same material was formalized in LEAN 4 and how one can increase confidence that the formal theorems match the intended informal claim.

2 Preliminaries

Throughout, $\mathbb{N}$ denotes the nonnegative integers $\{0, 1, 2, \dots\}$. Divisibility is written $d \mid n$.

Definition 2.1 (Prime). A natural number n is *prime* if $n > 1$ and every positive divisor of n is either 1 or n . In symbols,

$$\text{IsPrime}(n) \stackrel{\text{def}}{\iff} n > 1 \wedge (\forall d \in \mathbb{N}, d \mid n \Rightarrow d = 1 \vee d = n).$$

Definition 2.2 (Even). A natural number n is *even* if $2 \mid n$. In symbols,

$$\text{IsEven}(n) \stackrel{\text{def}}{\iff} 2 \mid n.$$

Remark 2.3. These are the usual textbook characterizations. Equivalently, $\text{IsEven}(n)$ if and only if $n \bmod 2 = 0$, and $\text{IsPrime}(n)$ if and only if $n > 1$ and n has no divisor d with $1 < d < n$. Both equivalences are routine and were checked in the companion formalization.

Lemma 2.4 (Two is an even prime). *We have $\text{IsPrime}(2)$ and $\text{IsEven}(2)$.*

Proof. Evenness is immediate: $2 = 2 \cdot 1$. For primality, the only positive divisors of 2 are 1 and 2, which one verifies by noting that any positive divisor d satisfies $d \leq 2$. $\square$

3 No even prime greater than two

We first prove the “upper” half of the story.

Theorem 3.1. *If $n > 2$ and $\text{IsEven}(n)$, then $\neg \text{IsPrime}(n)$. Equivalently: every even natural number greater than 2 is composite.*

We give five proofs. Each establishes the same mathematical content by a different organizing idea; all five have been checked independently in LEAN.

3.1 Proof A: Direct from the definition

Proof A. Let p be prime and even. Then $2 \mid p$. By [Definition 2.1](#) every positive divisor of p equals 1 or p , so $2 = 1$ or $2 = p$. The first alternative is absurd, hence $p = 2$. In particular, no even prime can exceed 2. $\square$

3.2 Proof B: Contrapositive

Proof B. Let $n > 2$ be even, and write $n = 2k$. Then $k > 1$ (since $n > 2$) and $k < n$ (since $k = n/2$). Thus k is a positive divisor of n strictly between 1 and n , so n is not prime. $\square$

3.3 Proof C: Contradiction

Proof C. Suppose p is an even prime with $p \neq 2$. Then $2 \mid p$, and primality forces $2 = 1$ or $2 = p$. Both are impossible under the standing assumptions, a contradiction. $\square$

3.4 Proof D: Prime divisibility

Lemma 3.2. *If p and q are prime and $p \mid q$, then $p = q$.*

Proof. By primality of q , the divisor p equals 1 or q . But $p > 1$, so $p = q$. $\square$

Proof D. Let p be an even prime. Then $2 \mid p$. Since 2 is itself prime ([Lemma 2.4](#)), [Lemma 3.2](#) yields $2 = p$. $\square$

3.5 Proof E: Least counterexample

Lemma 3.3 (Well-ordering form). *If a predicate P on $\mathbb{N}$ holds for some natural number, then there exists a least such number: some m with $P(m)$ and $\neg P(k)$ for all $k < m$.*

Proof E. Suppose there exists an even prime greater than 2, and let m be a least such number ([Lemma 3.3](#)). Then m is even and $m > 2$, so the argument of Proof B shows that m is composite—contradicting the choice of m . Hence no even prime greater than 2 exists. $\square$

Remark 3.4. For this particular theorem the appeal to minimality is logically redundant: *every* even integer greater than 2 is immediately composite. The least-counterexample packaging is included because it is a standard pattern, and because the well-ordering lemma itself is of independent interest.

Corollary 3.5. *If $\text{IsPrime}(p)$ and $\text{IsEven}(p)$, then $p = 2$.*

Proof. If $p > 2$, [Theorem 3.1](#) forbids primality. Since every prime satisfies $p > 1$, the remaining possibility compatible with evenness is $p = 2$. $\square$

4 No even prime less than two

The complementary “lower” half is even shorter, and in fact does not use evenness at all.

Theorem 4.1. *If $n < 2$, then $\neg \text{IsPrime}(n)$. In particular, there is no even prime less than 2.*

Proof. By Definition 2.1, every prime satisfies $n > 1$. On $\mathbb{N}$ the condition $n < 2$ is incompatible with $n > 1$. Explicitly, the only candidates are $n = 0$ and $n = 1$, neither of which is greater than 1. $\square$

Remark 4.2. One may equally argue by exhaustion on $\{0, 1\}$, or by quoting the characterization “ $\text{IsPrime}(n) \Rightarrow n > 1$ ” obtained from auditing Definition 2.1. All of these variants appear in the companion LEAN file.

5 The grand uniqueness theorem

We now assemble the two sides into a single expansive picture and deduce uniqueness.

Definition 5.1 (Expansive even-prime picture). The *expansive even-prime picture* is the conjunction of the following four assertions:

- (a) there is no even prime $n < 2$;
- (b) there is no even prime $n > 2$;
- (c) $\text{IsPrime}(2)$;
- (d) $\text{IsEven}(2)$.

Theorem 5.2 (Grand uniqueness). *The following are equivalent:*

- (i) *the expansive even-prime picture holds;*
- (ii) $\forall p \in \mathbb{N}, \text{IsPrime}(p) \rightarrow \text{IsEven}(p) \rightarrow p = 2$;
- (iii) *2 is an even prime, and it is the only one;*
- (iv) $\neg \exists p \in \mathbb{N} \text{ such that } \text{IsPrime}(p) \wedge \text{IsEven}(p) \wedge p \neq 2$.

In particular, there is no even prime aside from 2.

Proof. The implications among (ii)–(iv) are pure propositional rearrangements; we focus on the passage through the expansive picture.

$(i) \Rightarrow (ii)$. Assume the expansive picture, and let p be an even prime. By trichotomy on $\mathbb{N}$, exactly one of $p < 2$, $p = 2$, or $p > 2$ holds. The first case is ruled out by (a), the third by (b), and therefore $p = 2$.

$(ii) \Rightarrow (i)$. Clauses (c) and (d) are Lemma 2.4. Clause (a) is Theorem 4.1. For (b): if $n > 2$ is even and prime, then (ii) forces $n = 2$, contradicting $n > 2$.

$(ii) \Leftrightarrow (iii) \Leftrightarrow (iv)$. Immediate from Lemma 2.4 and first-order logic. $\square$

Corollary 5.3. *The unique even prime natural number is 2.*

6 On matching statements to proofs

Interactive theorem provers such as LEAN check a judgment of the form $\Gamma \vdash t : T$: under context Γ , the term t has type T . When T is a proposition, this is a formal proof of T . What the kernel *cannot* check is that T correctly renders an English sentence. This is the classical *specification gap* between verification and validation [1, 2].

6.1 What remains trusted

After formalization, a human must still accept a small *reading surface*. In our development that surface is essentially

```
def TargetClaim : Prop :=
  forall p : Nat, IsPrime p -> IsEven p -> p = 2
```

together with the definitions of IsPrime and IsEven from [Section 2](#). Everything else is derived.

6.2 What can be machine-checked

Inside the prover one can still compress that trust surface substantially:

1. **Independent restatements.** Formalize the intended claim in several equivalent ways (uniqueness; nonexistence of any other even prime; compositeness of all even $n > 2$) without referring to the proof theorems, then prove the restatements equivalent.
2. **Proof–claim alignment.** Show that each concrete proof theorem inhabits exactly the target proposition (and conversely).
3. **Definition audits.** Prove $\text{IsEven}(n) \leftrightarrow (n \bmod 2 = 0)$ and the “no proper divisor” characterization of primality, so that a silently wrong encoding is unlikely to survive.
4. **Anti-triviality.** Exhibit an even prime, an odd prime, and an even composite; show that the false claim “every even prime equals 3” is refuted. These tests catch vacuous or collapsed definitions.

In the companion development these steps are packaged as a single *specification certificate*, which the LEAN kernel accepts.

7 The Lean formalization

The accompanying LEAN 4 project uses only the standard library (no Mathlib), so that a bare toolchain suffices to check it. The module layout mirrors the mathematics of this note:

Module	Contents
EvenPrime/Prime.lean	IsPrime, IsEven, Lemma 2.4 , Lemma 3.2
EvenPrime/Proofs.lean	Five proofs of Theorem 3.1/Corollary 3.5
EvenPrime/BelowTwo.lean	Theorem 4.1 and variants
EvenPrime/Grand.lean	Expansive picture and Theorem 5.2
EvenPrime/Spec.lean	Specification certificate (Section 6)
EvenPrime/Tests.lean	Concrete computational checks

A representative fragment of the grand theorem reads as follows.

```

theorem no_even_prime_aside_from_two_of_expansive
  (H : ExpansiveEvenPrimePicture) :
  NoEvenPrimeAsideFromTwo := by
  obtain <hBelow, hAbove, _, _> := H
  intro p hp he
  by_cases hlt : p < 2
  . exact False.elim (hBelow p hlt he hp)
  . by_cases hgt : p > 2
    . exact False.elim (hAbove p hgt he hp)
    . omega

```

Concrete regression tests confirm, among other things, that 2 is even and prime while 4, 6, 8, 100 are even and not prime, and that odd primes such as 3, 5, 7 remain prime under the formal definition.

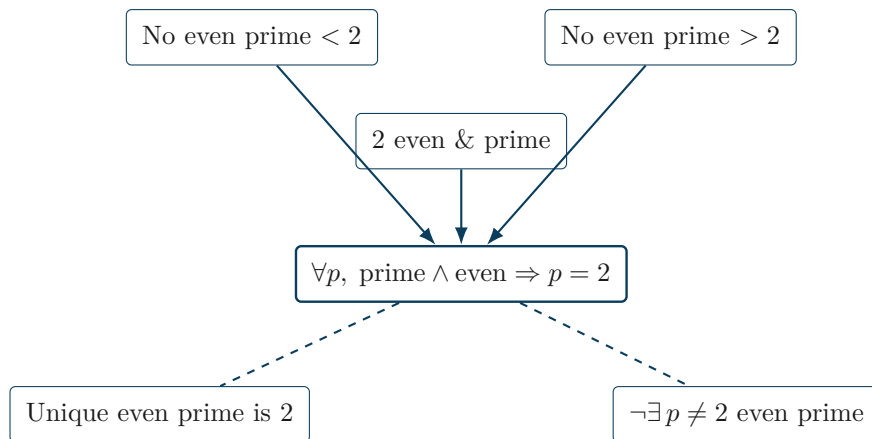
8 Conclusion

The uniqueness of the even prime is elementary, but it is also a tidy microcosm of mathematical practice: a crisp definition, several genuine proof ideas, a two-sided strengthening that makes uniqueness inevitable by trichotomy, and—once one formalizes—a reminder that the last inch of trust is always a human reading of a top-level statement. Within that discipline, the grand claim stands:

There is no even prime natural number aside from 2.

A Logical relationships at a glance

The following diagram summarizes the equivalences checked in the formalization. Solid arrows are the constructive implications used in the grand proof; dashed edges indicate propositional equivalences.



B Checklist for the trusted reading surface

A reader wishing to validate—rather than merely verify—the formalization need only confirm the following.

1. IsPrime and IsEven match [Definitions 2.1](#) and [2.2](#).
2. The target proposition is $\forall p, \text{IsPrime}(p) \rightarrow \text{IsEven}(p) \rightarrow p = 2$.
3. The expansive picture used in the grand proof is exactly [Definition 5.1](#).
4. The kernel accepts the development (e.g. via `lake build`).

References

- [1] R. A. De Millo, R. J. Lipton, and A. J. Perlis, *Social processes and proofs of theorems and programs*, Communications of the ACM **22** (1979), no. 5, 271–280.
- [2] A. Asperti, H. Geuvers, and R. Natarajan, *Social processes, program verification and all that*, Mathematical Structures in Computer Science **19** (2009), no. 5, 877–896.